

Fubswrjudskb

Fxuwlv Phqwrq

Wrgdb'v Wrslfv

- Orjlvwlfv
 - Fodvv olvw.
 - Vboodexv.
- Wkh Pdwk.
- Zkdw lv Fubswrjudskb.
- Vrrph Fodvvlfdo Fubswrvbvwhpv.

Cryptography

4003-482/4005-705

Curtis Menton

Today's Topics

- Logistics
 - Class list.
 - Syllabus.
- The Math.
- What is Cryptography.
- Some Classical Cryptosystems.

About Me



- BS/MS in CS at RIT, 2004-2009
- PhD in CS at UR, 2010-2013
 - Defended April 8
- Starting at Google in Mountain View in September
- Besides CS, I like photography, cycling, playing music (ukulele, banjo, guitar)

The Math

This is not a course that will simply describe a bunch of cryptographic algorithms and ways to break them.

You will see a lot of math and CS theory in this course, including,

- some abstract algebra (number theory, groups, rings, fields),
- some linear algebra,
- some probability and information theory, and
- some complexity theory.

The book provides mathematical background, but it is important that you are comfortable with math.

What is Cryptography

Cryptography is the study of secure communication over insecure channels.

To help describe the protocols, we use the following terminology:

Alice wants to communicate with *Bob* over an insecure channel in such a way that an eavesdropper *Eve* cannot understand what is being said.

All cryptosystems described in Chapter 2 (and Chapter 4) are *private key* cryptosystems: Alice and Bob Agree on a *key* before they communicate securely.

What is Cryptography

The information Alice wants to send is the *plaintext*. Alice *encrypts* the plaintext, using the *key*, and sends the resulting *ciphertext* over the insecure channel to Bob. Bob then *decrypts* the ciphertext, using the key, and reconstructs the plaintext.

Since Eve doesn't know the key, she cannot reconstruct the plaintext. At least, that's the idea.....

Since, in a private-key cryptosystem the encryption and decryption keys are the same or easily computable from each other, private-key systems are also called *symmetric-key cryptosystems*.

Eve

In addition to trying to read the message, Eve could try to find the key, so that she will be able to read all messages encrypted by this key.

In this book, Eve will also play the role of the malicious active attacker (usually called *Mallory*). In this mode, Eve could corrupt Alice's message, or masquerade as Alice.

Some Cryptanalysis Terminology

Cryptanalysis is the process of attempting to compute the key.
The most common attack models are:

- *ciphertext only attack*
- *known plaintext attack*
- *chosen plaintext attack*
- *chosen ciphertext attack*

What's the weakest type of attack?

Cryptography can do more

In addition to providing confidentiality, cryptography provides tools for achieving the following other security goals:

- Authentication.
- Data Integrity.
- Non-repudiation.

Chapter 2

Classical Cryptosystems

We will look at some older cryptosystems and ways to break them. This is also a good time to sneak in some of the math from Chapter 3.

Some Conventions for this chapter

In text encryption, plaintext will be written in lowercase. CIPHERTEXT will be written in uppercase.

Spaces and punctuation will usually be omitted.

We will often identify the letters of the alphabet with integers $\{0, \dots, 25\}$.

Monoalphabetic Ciphers

The simplest cryptosystems are *monoalphabetic*: each letter is mapped to a unique letter, and messages are encoded letter-by-letter.

Examples are the shift cipher, the substitution cipher, and the affine cipher.

The shift cipher (and many other classical ciphers) are based on *modular arithmetic*.

Modular Arithmetic

Note: All this is overkill for the shift cipher, but we will use modular arithmetic, groups, rings, and fields a lot, and this is a good place to introduce some of it.

If a and b are integers, and m is a positive integer, we write $a \equiv b \pmod{m}$ (a is *congruent* to $b \pmod{m}$) if m divides $a - b$.

True or false?

- $7 \equiv 5 \pmod{3}$
- $7 \equiv 1 \pmod{3}$
- $66 \equiv 0 \pmod{3}$
- $4 \equiv 1 \pmod{3}$
- $-4 \equiv -1 \pmod{3}$
- $-8 \equiv 7 \pmod{3}$

If a is an integer, and m is a positive integer, $a \bmod m$ is the remainder when a is divided by m . The remainder is an integer between 0 and $m - 1$ inclusive.

- $8 \bmod 3 =$
- $63 \bmod 7 =$
- $3 \bmod 6 =$
- $0 \bmod 2 =$
- $1 \bmod 1 =$
- $-8 \bmod 3 =$
- $-63 \bmod 7 =$

Is $\bmod$ the same as $\%$ in C/C++/Java?

$\mathbb{Z}_m$

$\mathbb{Z}_m$ is the set $\{0, \dots, m-1\}$ with two operations: addition (modulo m) and multiplication (modulo m).

$\mathbb{Z}_m$ is a *commutative ring*, i.e., addition and multiplication (mod m) are closed, associative, and commutative, 0 is the additive identity, each element has an additive inverse, and the distributive property is satisfied.

If $m > 1$, $\mathbb{Z}_m$ is a commutative ring with identity.

Shift Cipher

The *key* κ is an element of $\mathbb{Z}_{26}$.

Encrypting one letter $x \in \mathbb{Z}_{26}$ works as follows:

$$x \mapsto (x + \kappa) \pmod{26}.$$

How would you decrypt?

$$x \mapsto$$

Note: For $\kappa = 3$, this is known as the *Caesar cipher*, attributed to Julius Caesar.

Note: Of course, you can define the shift cipher over any $\mathbb{Z}_m$.

Usefulness of the Shift Cipher

The encryption and decryption functions are efficiently computable. That's good.

But the shift cipher (applied to a string) is not secure, not even against a ciphertext only attack. The cipher can be cryptanalyzed by *exhaustive key search*, since there are only 26 keys.

This assumes that Eve knows which cipher is being used. This is the general assumption in cryptanalysis: It is known as *Kerckhoff's Principle*.

How about the other types of attack?

Substitution Cipher

The substitution cipher is a monoalphabetic cipher that is defined by a permutation of the alphabet.

For example, suppose the key is

```
abcdefghijklmnopqrstuvwxyz  
ONETWHRFUISXVGABCDJKLMPQYZ
```

Decode the following ciphertext, using the above key:

```
EDYBKARDOBFY
```

Cryptanalysis of the Substitution Cipher

In general, exhaustive key search is not possible, since there are a huge number of keys. How many?

(Of course, if you know that a lazy person picked the key, that may reduce the key space considerably).

Still, the substitution cipher is easy to break (even with a ciphertext only attack), using statistical properties of the English language: look at the frequency of letters (Table 2.1, page 17), digrams (2 consecutive letters), and trigrams (3 consecutive letters). (See Section 2.4.)

Affine Cipher

A third monoalphabetic cryptosystem.

The key consists of a pair $(\alpha, \beta) \in \mathbb{Z}_{26} \times \mathbb{Z}_{26}$ such that $\gcd(\alpha, 26) = 1$.

Encrypt a letter x using the following *affine function*:

$$x \mapsto (\alpha x + \beta) \pmod{26}.$$

Decrypt as follows:

$$x \mapsto \alpha^{-1}(x - \beta) \pmod{26}.$$

Note: Of course, you can define the affine cipher over any $\mathbb{Z}_m$.

Questions about the Affine Cipher

How does the affine cipher relate to the shift cipher and the substitution cipher?

How many possible keys are there?

Is the affine cipher secure?

Why do we have the condition: $\gcd(\alpha, 26) = 1$?

What is α^{-1} ?

Are the encryption and decryption functions efficiently computable?

How would you cryptanalyze the affine cipher?